

SECUREBOX

Complete Hardware & Software Requirements

Portable Adaptive Security Gateway + Zero-Trust Enterprise Workspace + BYOD Platform

Version 1.0 - Prototype to Commercial Product Requirements

Core product promise: Plug in. Connect. Authenticate. Work securely.

Important: SecureBox should be designed to reduce attack surface and enforce security controls. It must not be marketed as impossible to hack or as guaranteeing that data can never be copied.

1. Product Definition

SecureBox is a portable hardware security gateway that protects devices on untrusted networks and provides a hardware-backed path to controlled enterprise resources and remote workspaces. It combines personal network protection with enterprise secure-workspace access.

Primary capabilities

- Secure VPN gateway for public, hotel, airport, home and office networks.
- Firewall, network isolation and VPN kill switch.
- Secure DNS and optional malicious-domain filtering.
- Rogue access point / evil-twin detection and network fingerprinting.
- IDS/IPS and network anomaly monitoring.
- Dynamic network risk scoring.
- Hardware-backed device identity.
- User authentication, MFA and enterprise identity integration.
- Adaptive policy engine using network, device, user and resource trust.
- Enterprise BYOD access without manual VPN/VDI configuration.
- Browser-based remote workspace, including Windows virtual desktops.
- Controlled workspace data policies and DLP.
- Central cloud management, audit logs and remote device revocation.

2. System Architecture

```
INTERNET
|
PUBLIC / HOME / OFFICE
|
v
+-----+
| SECUREBOX |
+-----+
| Network detection |
| Risk engine |
| VPN |
| Firewall |
| DNS security |
| IDS / IPS |
| Device trust |
| Policy engine |
| Hardware identity |
+-----+
|
Secure tunnel
|
v
+-----+
| SECURE CLOUD |
+-----+
| Identity |
| Policy |
| DLP |
| Windows VM / VDI |
| Company apps |
| Audit / logging |
+-----+
|
v
User's Mac / Windows /
```

The physical SecureBox should normally act as the secure gateway and trust anchor. A full Windows VM should run in the company datacenter or cloud rather than on the small physical gateway.

3. Prototype Hardware Requirements

Use off-the-shelf hardware for the first proof-of-concept. Do not design a custom PCB until the network architecture and workspace model are validated.

Component	Recommended prototype	Purpose
Main computer	Raspberry Pi 5, 8 GB	Gateway, routing, security services
Storage	128-256 GB SSD	OS, logs, configuration and test data
Wi-Fi	Built-in + optional USB Wi-Fi	WAN/LAN separation and scanning
Ethernet	Gigabit Ethernet	Stable WAN/LAN connection
Second network interface	USB Ethernet or second Wi-Fi	Separate untrusted and protected sides
Power	USB-C supply	Portable power
Security chip	Optional TPM 2.0 / secure element	Hardware-backed identity
Cooling	Active cooling	Stable VPN/IDS performance
Enclosure	3D-printed prototype case	Portable form factor
Optional cellular	USB 4G/5G modem	Internet where Wi-Fi is unavailable
Optional display	Small OLED	Status information

Recommended network topology

External / untrusted side -> SecureBox -> Protected side
Public Wi-Fi / Ethernet | Secure Wi-Fi / Ethernet
+> VPN / Firewall / DNS / IDS

4. Commercial Hardware Requirements

Area	Target requirement
Processor	Embedded SoC/CPU selected for encrypted throughput, routing, IDS/IPS and local policy evaluation
RAM	Target 4-8 GB minimum depending on IDS/IPS and local services
Storage	32-128 GB flash/eMMC minimum; more if local logs are retained
Wi-Fi	Wi-Fi 6/6E or newer, subject to cost and certification
Ethernet	At least 1 GbE; premium model may use 2.5 GbE
USB	USB-C for power/data; optional USB-A/USB-C peripherals
Security	Secure element and/or TPM-class hardware-backed key storage
Cellular	Optional 4G/5G modem
Boot security	Secure boot and signed firmware
Updates	Authenticated, signed, rollback-safe firmware
Physical design	Portable, thermally managed, tamper-aware enclosure

5. Prototype Software Requirements

Layer	Recommended technology	Role
Gateway OS	OpenWrt or Linux	Routing, firewall, wireless and network services
VPN	WireGuard	Encrypted tunnel
Firewall	nftables / OpenWrt firewall	Filtering, NAT, kill switch and isolation
DNS	Unbound / dnsmasq + DoH/DoT	Secure DNS and policy filtering
IDS/IPS	Suricata	Traffic inspection and detection
Network tools	iw, tcpdump, hostapd, iproute2	Scanning, capture, routing and AP functions
Risk engine	Python initially	Network fingerprinting and adaptive scoring
Backend	Node.js + TypeScript	API, device management and policy
Database	PostgreSQL	Users, devices, policies and events
Admin UI	React / Next.js / TypeScript	Enterprise administration
Authentication	OIDC/SAML, MFA, passkeys	Enterprise identity
Workspace	Windows VM / VDI	Remote Windows environment
Remote access	Browser-based remote desktop	No-client/low-client workspace
Containers	Docker	Repeatable development environments
Source control	Git	Versioning

6. Network Security Software Requirements

VPN

WireGuard is a strong prototype choice. SecureBox should support automatic tunnel establishment, key management, route control and a VPN kill switch. If the tunnel fails, traffic should not silently fall back to an unprotected network when policy requires VPN-only access.

Firewall

- Stateful filtering
- Block unsolicited inbound traffic
- LAN/client isolation
- Port and protocol policy
- VPN-only routes where required
- Per-device policy
- Emergency deny mode

DNS security

- Protected DNS resolver
- Optional DNS-over-HTTPS / DNS-over-TLS
- Malicious/phishing-domain blocking
- Enterprise allowlist/blocklist
- DNS anomaly logging

IDS/IPS

- Suricata or equivalent

- Port-scan detection
- Known attack signature detection
- Suspicious connection-pattern detection
- Security event generation

7. Network Risk and Adaptive Security

The risk engine should not claim that a network is absolutely safe or unsafe. It generates an assessment used to select a security policy.

Signal	Example contribution
Network fingerprint	Compare observed characteristics with known/expected characteristics
Gateway behavior	Unexpected gateway changes or unusual responses
DHCP behavior	Unexpected configuration or inconsistent parameters
DNS behavior	Unexpected resolver or suspicious responses
ARP behavior	Unexpected gateway MAC changes or anomalies
Rogue AP indicators	SSID/BSSID and behavior inconsistencies
Threat intelligence	Known malicious infrastructure indicators
Historical behavior	Changes from prior trusted observations

Example policy levels

Level	Typical behavior
Trusted	Normal firewall; VPN according to policy; protected DNS
Untrusted	VPN mandatory; strict firewall; LAN isolation; restricted inbound traffic
Dangerous	Maximum firewall; full isolation; VPN mandatory; restricted corporate workspace access

8. Trust Model

```

NETWORK TRUST
|
v
+-----+
| POLICY |
| DEVICE TRUST -> | ENGINE | <- USER TRUST
+-----+
|
v
SECURITY / DATA POLICY

```

The policy engine can consider network trust, device trust, user identity, resource sensitivity, company policy, time and context. The result determines VPN requirements, firewall profile, LAN isolation, workspace access and DLP level.

Hardware identity

Each SecureBox should have a unique cryptographic identity. A secure element or TPM-class component should protect private keys. The cloud should register, authenticate, suspend and revoke individual devices.

9. Enterprise BYOD Requirements

The key enterprise scenario is allowing an employee to use a personal Mac, Windows PC, Chromebook or other supported device while accessing company-controlled resources through SecureBox.

```
Personal laptop / phone
|
v
SecureBox
|
Hardware identity
|
User authentication
|
Network risk evaluation
|
Adaptive policy
|
Secure tunnel
|
Company cloud
|
Remote workspace / apps
```

No-install / low-install objective

The preferred experience is to avoid manual VPN and VDI configuration. The user connects to SecureBox, opens a browser, authenticates and launches the secure workspace. Advanced endpoint controls may require an optional agent; zero software cannot guarantee every host-level control.

10. Windows / Cross-Platform Workspace Requirements

A user with a Mac can receive a Windows desktop through SecureBox. The Windows OS should normally run remotely in a company datacenter or cloud.

```
MacBook
|
Browser
|
SecureBox portal
|
Authentication
|
Secure tunnel
|
Windows VM / VDI
|
Windows applications
```

Supported physical endpoints

- macOS laptop
- Windows laptop
- Linux laptop
- Chromebook
- Compatible tablet or phone for selected workspace use cases

Workspace technology options

- Windows virtual machines
- Enterprise VDI platforms
- Browser-based remote desktop gateway
- Remote application streaming
- Company-hosted or cloud-hosted workspaces

11. Data Protection and DLP Requirements

Sensitive data should preferably remain in the controlled remote workspace rather than being downloaded to the personal computer.

Control	Example policy
Download	Blocked for confidential workspace
Upload	Restricted to approved destinations
Clipboard	Blocked or restricted
Printing	Blocked or restricted
Local drive mapping	Blocked
USB transfer	Restricted where workspace platform supports it
File sharing	Restricted
External cloud storage	Blocked or restricted
Session logging	Record policy-relevant events subject to privacy/legal requirements

Important limitation: a gateway cannot guarantee that a determined person cannot copy information from an unmanaged endpoint. The strongest architecture minimizes local possession of sensitive data.

12. Cloud / Backend Requirements

Service	Responsibility
API server	Device, user, policy and workspace APIs
Identity service	Authentication, SSO, MFA and sessions
Device service	Registration, certificates, status and revocation
Policy service	Security and data-access policies
PostgreSQL	Structured application data
Event pipeline	Security and audit events
Workspace service	Windows VM / VDI assignment and lifecycle
Admin portal	Company IT management
Monitoring	Health, uptime and security telemetry
Notification service	Security alerts and administrative events

13. Database Requirements

Prototype PostgreSQL tables can include:

- organizations
- users
- devices
- device_credentials
- device_assignments
- policies
- network_events
- security_events
- sessions

- workspaces
- workspace_assignments
- audit_logs
- firmware_versions

The schema should support organization isolation, device revocation, policy versioning and auditable changes.

14. Admin Dashboard Requirements

```

SECUREBOX ADMIN
-----
Devices 1,245
Online 987
High Risk 18
Security Events 142

Device Risk Status
SB-1842 HIGH ONLINE
SB-3911 LOW ONLINE
SB-4412 MEDIUM OFFLINE

Actions:
[View] [Change Policy] [Revoke] [Open Audit]
```

Administrative capabilities

- Register and assign devices
- Assign users and policies
- Set network security profiles
- Set DLP/workspace policies
- Revoke lost or compromised devices
- View risk and security events
- View audit history
- Manage firmware versions
- Integrate with enterprise identity providers

15. User Experience Requirements

```

CONNECT SECUREBOX
|
CONNECT TO SECUREBOX NETWORK
|
OPEN BROWSER
|
AUTHENTICATE
|
SECURITY CHECK
|
OPEN SECURE WORKSPACE
|
WORK
```

The user should not have to manually configure WireGuard, certificates, routes, DNS or firewall rules.

16. Mobile and Tablet Requirements

Phones and tablets can use SecureBox as a protected network gateway. They connect to SecureBox's private Wi-Fi instead of directly connecting to the untrusted public network.

- Android support
- iPhone/iPad support
- Protected Wi-Fi access
- VPN gateway protection
- DNS security
- Network isolation
- Optional companion app for status, alerts and account management

17. Device Security Requirements

- Secure boot
- Signed firmware
- Hardware-backed cryptographic keys
- Authenticated firmware updates
- Rollback protection where appropriate
- Credential revocation
- Encrypted sensitive configuration
- Minimal exposed management services
- Strong provisioning credentials
- Authentication rate limiting
- Tamper-aware enterprise design

18. Logging and Monitoring

Security-relevant events should be recorded with timestamps, device identity, policy context and outcome, while minimizing unnecessary personal data.

- Device connection/disconnection
- Network risk changes
- VPN establishment/failure
- Authentication success/failure
- Policy changes
- Workspace launch/termination
- DLP policy events
- Firmware changes
- Device revocation

19. Developer Workstation Requirements

Item	Recommended
Operating system	Windows 11, macOS or Linux
RAM	16 GB minimum; 32 GB preferred for VMs
Storage	100+ GB free SSD; more for Windows VM images

Item	Recommended
CPU	Modern multi-core CPU with hardware virtualization
Development tools	VS Code or equivalent, Git, Docker
Network lab	Separate test router/AP and test devices
Cloud account	Small development environment for API/database/workspace testing

20. Prototype Security Laboratory

```

INTERNET
|
TEST ROUTER
|
+-----+-----+
| |
SecureBox Test AP
|
+-----+-----+
| | |
Laptop Phone Tablet

```

Perform security testing only on equipment and networks that you own or are explicitly authorized to test.

21. Prototype Cost Planning

Component	Rough prototype range (INR)
Raspberry Pi 5	8,000-12,000
SSD	2,000-5,000
USB Wi-Fi / network adapter	1,000-3,000
Power supply	1,000-2,000
Secure element	500-1,500
Cooling / case	1,000-2,000
Cables / accessories	Around 1,000
Estimated total	About 14,500-26,500

Planning ranges only; not vendor quotations.

22. Development Roadmap

Phase	Goal	Main deliverable
1	Secure gateway	Private Wi-Fi, routing, firewall and Internet
2	VPN	WireGuard, kill switch and protected DNS
3	Risk engine	Network fingerprinting, scoring and adaptive profiles
4	Enterprise identity	User authentication and hardware-backed device identity
5	Cloud management	API, PostgreSQL and admin dashboard
6	Workspace	Browser-based Windows VM / VDI
7	DLP	Controlled workspace data-transfer policies
8	Custom hardware	Commercial PCB/enclosure and secure boot
9	Pilot	Controlled real-world enterprise validation

23. Patent / IP Considerations

Do not assume the overall concept is automatically patentable. VPN gateways, travel routers, zero-trust systems, VDI, DLP, device certificates and hardware security modules are established categories.

Potential areas for prior-art investigation

- Portable gateway that dynamically changes security policy from network trust.
- Hardware-backed SecureBox identity used as the trust anchor for remote workspace access.
- Combined trust engine using network, device, user and resource sensitivity.
- No-install or low-install enterprise workspace workflow enabled by a physical security gateway.
- One portable device that transitions between consumer network protection and enterprise zero-trust workspace mode.

Candidate invention statement for investigation

A portable hardware security gateway that establishes a hardware-backed trust relationship with an enterprise environment, evaluates network trust, endpoint trust and user trust, dynamically selects security and data-access policies, and provides controlled access to remote computing resources without requiring conventional endpoint security configuration.

This is a patent hypothesis, not a conclusion of novelty. A professional patent search and claim drafting process is required.

24. Security and Product Limitations

- Do not claim SecureBox is impossible to hack.
- Do not claim that no data can ever be copied.
- A gateway cannot fully control an unmanaged endpoint's screen, camera or physical actions.
- The strongest data-protection design keeps sensitive data in a controlled remote workspace.
- Endpoint agents may still be necessary for advanced host-level controls.
- Security scores are risk assessments, not proof of safety.
- Commercial deployment requires security testing, threat modeling and appropriate compliance review.

25. Final Build Checklist

Area	Requirement	Priority
Hardware	Raspberry Pi 5 + SSD + dual network capability	Must
Networking	Private LAN/Wi-Fi behind gateway	Must
Firewall	nftables/OpenWrt firewall	Must
VPN	WireGuard + kill switch	Must
DNS	Secure resolver and policy	Must
IDS/IPS	Suricata or equivalent	High
Risk engine	Network assessment and adaptive policy	High
Hardware identity	Secure element/TPM	High
Backend	Node.js/TypeScript API	High
Database	PostgreSQL	High

Area	Requirement	Priority
Admin UI	React/Next.js	High
Identity	OIDC/SAML + MFA	High
Workspace	Windows VM/VDI	High
Browser access	Remote workspace through browser	High
DLP	Download/clipboard/printing/file-transfer policy	High
Updates	Signed firmware updates	Later
Cellular	Optional 4G/5G	Later
Custom PCB	Commercial hardware	Later

26. Recommended First Demonstration

1. Connect SecureBox to a controlled public-style/test network.
2. SecureBox assesses the network.
3. Display a risk level based on controlled test conditions.
4. Establish VPN automatically.
5. Activate firewall and LAN isolation.
6. Connect a MacBook to SecureBox.
7. Open a browser and authenticate.
8. Launch a Windows workspace remotely.
9. Keep test confidential data inside the controlled workspace.
10. Revoke the device from the admin portal and verify workspace access stops.

If these steps work reliably, the project has a strong proof-of-concept foundation.

27. Final Product Vision

SecureBox should feel simple while the underlying architecture is sophisticated: a portable security boundary that travels with the user, protects devices on untrusted networks, provides hardware-backed enterprise identity, dynamically adapts security policy, and gives employees access to controlled computing environments from their existing hardware.

Product message: Plug in. Connect. Authenticate. Work securely.